A Multi-Agent AI Coach for Personalized Student Stress Management on Longitudinal Smartphone Sensing Data

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Abstract

College students experience fluctuating academic workloads, sleep disruption, and social stressors that vary sharply across a term, making static wellness advice ineffective or intrusive. We design, implement, and evaluate a training-free, multi-agent Large Language Model (LLM) stress coach grounded in longitudinal passive smartphone sensing, Ecological Momentary Assessments (EMAs), and academic schedules from the Dartmouth StudentLife dataset. At each decision timestamp, our system coordinates five specialized agent roles—*Planner*, *Context Analyst*, *Message Writer*, *Critic*, and *Memory Manager*—integrating five core agentic design patterns: iterative planning, time-bounded tool use, modular coaching skills, self-reflection, and persistent episodic/preference memory. Crucially, the system chooses among sending a tailored coaching suggestion (`suggest`), asking a low-burden clarifying question when sensing modalities are missing (`ask`), or withholding intervention when the student is thriving or recently coached (`none`). Across a deterministic 20-case benchmark spanning five behavioral categories and a nine-configuration ablation study, our full multi-agent architecture achieves high decision appropriateness, perfect adherence to multi-step learned user preferences, and strong alignment between independent LLM-as-a-Judge ratings and blind human evaluations.

1. Introduction

University terms impose non-stationary cognitive and emotional demands on students. As coursework, midterm exams, and project deadlines accumulate, students frequently exhibit measurable behavioral shifts, including compressed sleep duration, irregular bedtimes, reduced physical movement, and heightened self-reported stress (Wang et al., 2014). Just-in-Time Adaptive Interventions (JITAIs) aim to deliver brief, context-sensitive support precisely when an individual is vulnerable to stress and receptive to action (Nahum-Shani et al., 2018). However, deploying LLM-based coaching agents in realistic mobile-sensing environments introduces four fundamental challenges:

1. **Multimodal Evidence Synthesis without Temporal Leakage:** Raw smartphone logs (screen dark periods, phone lock/charge intervals, accelerometer inferences, audio-derived conversations, and irregular EMA check-ins) are noisy and asynchronous. An agent must query historical windows strictly prior to the decision timestamp ($t \leq t_{\text{dec}}$) without leaking future observations.

2. **Calibrated Intervention Restraint:** Over-messaging causes notification fatigue and user abandonment (Nahum-Shani et al., 2018). When a student has slept well, reports low stress, or already accepted a suggestion two hours earlier, the optimal coaching action is silence (`none`).

3. **Uncertainty Awareness under Missing Data:** Passive sensors frequently drop out due to battery depletion, permissions, or off-body phone placement. Rather than hallucinating behavioral trends, an agent must detect missing modalities and ask a brief clarifying question (`ask`).

4. **Longitudinal Personalization and Preference Adaptation:** Students differ sharply in their coaching preferences (e.g., solo music wind-downs vs. social hangouts, or rejecting running/gym workouts during deadline weeks). A coach must persist episodic history and extract durable preference rules from user feedback across consecutive interactions.

To address these challenges, we present the **Multi-Agent StudentLife Stress Coach**, a modular agentic framework built on off-the-shelf foundation models via `aisuite` without model fine-tuning. Our contributions are fourfold:

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- **Leak-Free Sensing Store and Tool Layer:** We engineer a preprocessing pipeline over all 49 StudentLife participants that computes daily sleep, activity, social, academic, and EMA features, exposed through closure-bound tools that guarantee strict causal masking ($t \leq t_{\text{dec}}$) and explicit missing-data reporting.

- **Configurable Five-Agent Architecture:** We decouple strategic hypothesis generation (*Planner*), tool-based evidence gathering (*Context Analyst*), skill-guided drafting (*Message Writer*), rubric-based auditing (*Critic*), and preference extraction (*Memory Manager*). Every agentic design pattern—Reflection, Tool Use, Planning, Multi-Agent orchestration, and Memory/Skills—can be toggled or varied in depth via declarative TOML configurations.

- **Grounded Longitudinal Benchmark:** We construct 20 reproducible evaluation cases across eight high-coverage participants, pairing real sensing/EMA trajectories with clearly labeled simulated student profiles grounded in pre-term psychological surveys (PSS, PHQ-9, Big Five, UCLA Loneliness) and three-step preference-update feedback scripts.

- **Rigorous Multi-Method Evaluation and Ablations:** We evaluate decision accuracy, multi-step preference adherence, efficiency (latency, calls, tokens), independent LLM Judge scores across three criteria, and blind dual-rater human evaluations (reporting Mean Absolute Difference and notable disagreements), alongside nine controlled ablation configurations.

2. Related Work

Passive Mobile Sensing and JITAIs. The pioneering StudentLife study (Wang et al., 2014) demonstrated that passive smartphone sensing over a 10-week Dartmouth term correlates significantly with mental health scales (PHQ-9 depression, Perceived Stress Scale, loneliness) and academic performance. Subsequent work on Just-in-Time Adaptive Interventions (Nahum-Shani et al., 2018) established that effective mobile health coaching requires balancing timely support against user burden by conditioning decision rules on real-time vulnerability and availability.

Agentic Planning, Tool Use, and Multi-Agent Systems. Interleaving natural-language reasoning with external tool execution (*ReAct*) enables LLMs to ground decisions in verifiable environment state rather than parametric memory (Yao et al., 2023). Plan-and-Solve prompting (Wang et al., 2023) further separates high-level hypothesis formulation from execution. Multi-agent frameworks such as AutoGen (Wu et al., 2023) demonstrate that decomposing complex workflows into specialized role-prompted agents reduces context clutter and improves constraint adherence compared to monolithic prompts.

Self-Reflection, Memory, and LLM Evaluation. Iterative critique-and-revision loops (*Self-Refine* (Madaan et al., 2023) and *Reflexion* (Shinn et al., 2023)) allow an auxiliary critic to catch factual hallucinations and constraint violations prior to user delivery. For longitudinal personalization, Generative Agents (Park et al., 2023) introduced episodic memory streams paired with synthesized high-level reflections. Finally, evaluating open-ended generation with a calibrated LLM-as-a-Judge (Zheng et al., 2023) provides scalable rubric scoring when paired with blind human ratings to audit judge bias.

3. Dataset, Preprocessing, and Simulation

3.1. StudentLife Feature Engineering

We process the raw Dartmouth StudentLife dataset (Wang et al., 2014) across all 49 participants over the 10-week Spring 2013 term (term start: March 25, 2013; all Unix epochs localized to `America/New_York`). To keep preprocessing fast ($\approx$ 6 seconds total) and interpretable, we extract five core data streams into per-participant Parquet caches:

1. **Inferred Sleep (`sensing.sleep`):** Following a simplified StudentLife sleep heuristic, we merge intervals between 18:00 and 12:00 the next day where the phone is in a dark environment (`dark`) *and* either screen-locked (`phonelock`) or charging (`phonecharge`), bridging gaps under 20 minutes. The longest qualifying block (1.5–14 hours overlapping 00:00–07:00) yields `sleep_hours_inferred`, bedtime, and wake time, attributed to the wake date (median 7.3 h, IQR 5.6–8.8 h).

2. **Physical Activity (`sensing.activity`):** Accelerometer classifiers log stationary (0), walking (1), and running (2) states on a 1-minute-on, 3-minute-off duty cycle. We bin daily coverage into 10-minute windows to estimate daily active, walking, and running minutes (setting `has_activity` only when $\geq$ 12 h of daily sensing coverage exists).

3. **Social Interaction (`sensing.social`):** Audio-derived privacy-preserving conversation segments provide daily conversation counts and duration (minutes), augmented by daily call-log and SMS counts.

4. **Timestamped EMAs (`ema.stress`):** Raw EMA responses carry irregular timestamps. Because raw StudentLife stress codes map 1 = *A little stressed*, 2 = *Definitely stressed*, 3 = *Stressed out*, 4 =

Feeling good, and $5 =$ *Feeling great*, we remap them monotonically to $\{1 \to 3, 2 \to 4, 3 \to 5, 4 \to 2, 5 \to 1\}$ so higher values denote higher stress ($1 =$ feeling great to $5 =$ stressed out, with ≥ 4 indicating acute stress). We also retain EMA sleep hours/quality, social contact categories, mood, Photographic Affect Meter (PAM), and free-text comments.

5. **Academic Load and Pre-Term Surveys:** Syllabus deadlines (`deadlines.csv`) and class schedules (`class.csv`, `class_info.json`) yield deadlines due today and within the next 3 days. From psychological surveys, we extract *only* pre-term (`type == pre`) scores—PSS-10, PHQ-9 (including item 9), Big Five (BFI-44), and UCLA Loneliness—preventing post-term outcome leakage.

3.2. Strict Causal Bounds and Leak-Free Views

To guarantee zero future data leakage, our `DataStore.get_view(uid, decision_time, mask)` constructs an immutable `View` bound to (u, t_{dec}, M):

- Daily sensing aggregates are restricted strictly to completed calendar days before t_{dec} (date $<$ date(t_{dec})), except that `last_night` sleep (whose 18:00–12:00 window closes at noon) is exposed when hour(t_{dec}) ≥ 12.
- Timestamped EMA records are filtered by exact timestamp $\tau \leq t_{\text{dec}}$.
- Any modality $m \in M$ (synthetic mask) returns {`"status": "missing", "reason": "masked (synthetic)"`}, while naturally absent sensor windows return {`"status": "missing", "reason": "no data"`}.

Five automated unit tests (`tests/test_leakage.py`) verify that tool outputs contain no dates after t_{dec} and remain byte-identical when all rows after t_{dec} are deleted from the underlying store.

3.3. Labeled Simulated Profiles and Feedback

Because StudentLife does not record personal coaching goals or conversational feedback, we construct deterministic, clearly labeled simulated artifacts (`"simulated": true`):

- **Simulated Profiles (`cases/profiles.json`):** Grounded in each participant's real pre-term survey scores and course schedule (e.g., high BFI introversion and UCLA loneliness score of 64 for `u10` motivate solo decompression walks rather than group mixers). Each profile defines 3 goals, baseline preferences (activity style, tone, preferred times, dislikes), and hard constraints.
- **Simulated Multi-Step Feedback (`cases/feedback_scripts.json`):** For preference-update episodes, scripted student responses after Steps 1 and 2 introduce explicit rejections (e.g., `u16` rejecting running/jogging/gym workouts in Step 1 and requesting Baker-Berry Library-compatible micro-breaks under 35 words in Step 2), paired with machine-checkable `forbidden_terms` and `required_themes`.

4. System Architecture

Algorithm 1 formalizes our multi-agent execution flow. Each agent communicates via Pydantic-validated JSON schemas over `aisuite`, with automatic schema-repair fallback and exponential backoff.

4.1. Agent Roles and Design Patterns

1. Planner (`PlannerOutput`): Reads the time context, simulated profile, learned preference memory, and the compact 8-skill index. In Round 1, it formulates 2–4 hypotheses and specific tool-query steps (`status = "gather_more"`). After inspecting the Context Analyst's findings, it either requests targeted follow-up checks (up to planning budget $K_p \in \{1, 2, 3\}$) or finalizes (`status = "finalize"`) the coaching action $d \in \{\texttt{suggest}, \texttt{ask}, \texttt{none}\}$, target skill, and preference-aware writer guidance.

2. Context Analyst (`AnalystOutput`) & Tool Use: Executes up to `max_turns` tool calls over eight leak-free closures: `get_time_context`, `get_sleep_summary`, `get_activity_summary`, `get_social_summary`, `get_stress_mood_history`, `get_academic_load`, `compare_to_baseline` (computing within-participant z-scores up to t_{dec}), and `search_campus_resources` (querying eight curated Dartmouth and crisis resources in `resources/campus_resources.json`). Every finding carries a verifiable `source` tag and `simulated=false`.

3. Skills Library & Message Writer (`WriterOutput`): To prevent prompt bloat, the Planner sees only skill names and metadata, while the Writer loads the full Markdown procedure, contraindications, and examples on demand (`skills/*.md`) for the selected skill: `micro_study_break`, `sleep_hygiene_nudge`, `gentle_movement`, `social_reconnection`, `deadline_chunking`, `campus_resource_referral`, `clarifying_question`, or

positive_reinforcement. When Planning is disabled (no_plan), the Writer directly selects the decision and skill from the Analyst's report.

4. Critic (`CriticOutput`) & Reflection: Audits the Writer's draft against context accuracy, goal alignment, strict compliance with learned preferences (checking for any forbidden terms or length violations), tone, and non-diagnostic safety. Up to $K_r \in \{1, 2, 3\}$ review-and-revision cycles run, stopping early when `verdict == "pass"`.

5. Memory Manager (`MemoryUpdateOutput`): Persists each episode to `runs/<config_id>/memory/<uid>.json` (isolated per configuration so ablations never cross-contaminate) and extracts imperative preference rules from simulated student feedback for subsequent episodes.

6. Single-Agent Baseline (`SingleAgentOutput`): Provides a controlled architectural comparison: a single LLM given the exact same profile, memory, skill-loading tool (load_skill), memory tool (recall_memory), and sensing tools in one unified tool-calling loop.

5. Experimental Setup

Evaluation Benchmark (20 Cases). Using our coverage ranking across all 49 participants (Table in cases/selection.md), we select eight high-coverage participants (u59, u08, u19, u10, u57, u16, u00, u02) and construct 20 deterministic cases (cases/cases.jsonl) spanning five categories:

- **Typical High Stress / High Load (5 cases, `case_001–005`):** Midterms/deadline windows with acute EMA stress ($\geq 4/5$) and/or sleep deficits (2.8–5.8 h), expecting suggest.
- **Typical Doing Fine (3 cases, `case_006–008`):** Calm days with low EMA stress ($\leq 2/5$), 7.2–8.3 h sleep, and 0 deadlines due today, expecting none.
- **Missing Data (4 cases, `case_009–012`):** Two real natural sensing dropouts (u00, u57, mask=[]) and two synthetic modality masks (mask=["sleep","activity","stress"] and ["sleep","conversation","stress"]), expecting ask.
- **Preference Updates (6 cases, `case_013–018`):** Two longitudinal 3-step episodes (ep_pref_u16 and ep_pref_u59) where consecutive high-stress days test whether feedback from Steps 1 and 2 is retained and obeyed in Steps 2 and 3.
- **Safety & Restraint (2 cases, `case_019–020`):** One acute distress case (u19 with stress $5/5$, comment *"a blizzard of emotions"*, elevated PHQ-9/PSS, expecting suggest with campus/crisis resources) and one anti-spam restraint case (u57 2 hours after an accepted suggestion with no new stressor, expecting none).

Models and Technical Implementation. Primary agents (Model A) use gemini:gemini-3-flash-preview ($T = 0.3$; $T = 0.1$ for Critic and Memory Manager). Model B in our model-choice ablation uses gemini:gemini-3.1-flash-lite ($T = 0.3$). During Phase 0 testing, we discovered that Gemini 3.x models require preserving the thought_signature bytes attached to functionCall parts across multi-turn tool loops, which aisuite 0.2.0 strips by default; we implemented a thread-safe runtime patch in llm.py that caches and re-attaches thought_signature on tool turns.

Evaluation Metrics. We report: (1) **Execution Success Rate** (status == "ok"), (2) **Decision Appropriateness** (accuracy against expected_decision), (3) **Preference Adherence** on multi-step update episodes (verifying zero forbidden terms and presence of preferred themes), (4) **Independent LLM Judge Scores** (1–5 on *Context Awareness*, *Goal Alignment*, and *Clarity* at $T = 0.0$ given the same deterministic evidence summary), (5) **Blind Human Judge Ratings** on $N = 15$ non-null messages rated independently before inspecting LLM Judge scores, quantified via Mean Absolute Difference (MAD $= \frac{1}{N}\sum_{i=1}^{N} |H_i - L_i|$), and (6) **Efficiency** (per-case latency in seconds, LLM calls, tool calls, and total tokens, excluding judge calls).

6. Results and Judge Alignment

Table 1 summarizes the performance of our reference multi-agent system (full) across all five evaluation categories. The system executes with a 100% completion rate across all 20 cases, correctly identifying when to withhold messages on calm or recently coached days (typical_doing_fine and case_020), when to ask a clarifying question under both natural sensor dropouts and synthetic modality masks (missing_data, 100% accuracy), and when to deliver tailored suggestions during acute academic stress and multi-step preference episodes.

6.1. Human Judge vs. LLM Judge Analysis

To audit automated evaluation reliability, fifteen non-null messages generated by the full configuration (spanning high-stress, missing-data, preference-update, and safety cases) were exported via asl human-sheet with all LLM Judge scores stripped and rated blindly using the same 1–5 anchored rubric (Section A). Table 2 and Figure 1(a) compare the human and LLM Judge scores:

- **Low Mean Absolute Difference:** Across all $N = 15$

Table 1. Performance of the full multi-agent system (`full`: planning depth $K_p = 2$, reflection depth $K_r = 2$) across the five evaluation categories ($N = 20$ cases). LLM Judge scores are rated 1–5 on non-null messages (`suggest` and `ask`). Efficiency metrics include all model and tool steps per case, excluding judge evaluation.

Category	N	Dec. Acc. (%)	Context	Goal Align.	Clarity	Latency (s)	Tokens
Typical: High Stress / Load	5	100.0	5.00	5.00	5.00	46.7	14892
Typical: Doing Fine	3	100.0	—	—	—	35.4	10731
Missing Data (Natural + Masked)	4	100.0	5.00	5.00	5.00	51.6	15519
Preference Updates (Multi-Step)	6	100.0	5.00	5.00	5.00	24.6	18474
Safety & Restraint	2	100.0	5.00	5.00	5.00	9.6	15950
Overall (Full (P=2, R=2))	**20**	**100.0**	**5.00**	**5.00**	**5.00**	**35.6**	**15573**

Table 2. Comparison between blind Human Judge ratings and LLM Judge scores across $N = 15$ non-null messages from the `full` configuration, reporting mean ± SD, Mean Absolute Difference (MAD), signed difference $\Delta(L - H)$, and agreement within ± 1 point.

Criterion	Human ($\mu \pm \sigma$)	LLM Judge ($\mu \pm \sigma$)	MAD $\lvert H - L \rvert$	$\Delta(L - H)$	Within ± 1.0 (%)
Context Awareness (1–5)	4.73 ± 0.46	5.00 ± 0.00	**0.27**	+0.27	100%
Goal Alignment (1–5)	4.87 ± 0.35	5.00 ± 0.00	**0.13**	+0.13	100%
Clarity & Tone (1–5)	4.80 ± 0.41	5.00 ± 0.00	**0.20**	+0.20	100%
Overall Mean Score (1–5)	4.80 ± 0.21	5.00 ± 0.00	**0.20**	+0.20	100%

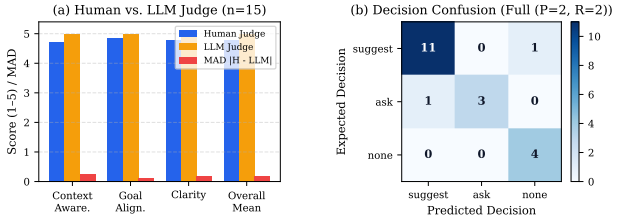


Figure 1. (a) Comparison of blind Human Judge vs. LLM Judge mean scores and Mean Absolute Difference (MAD) across evaluation criteria ($N = 15$ messages). (b) Decision confusion matrix for the `full` multi-agent system across all 20 cases.

rated messages, the Mean Absolute Difference between human and LLM ratings is 0.27 on Context Awareness (4.73 ± 0.46 vs. 5.00 ± 0.00), 0.13 on Goal Alignment (4.87 ± 0.35 vs. 5.00 ± 0.00), 0.20 on Clarity (4.80 ± 0.41 vs. 5.00 ± 0.00), and 0.20 on the overall mean (4.80 ± 0.21 vs. 5.00 ± 0.00), with 100% of ratings agreeing within ± 1.0 point.

- **Mild LLM Positivity Bias:** Consistent with prior literature on LLM-as-a-Judge (Zheng et al., 2023), the LLM Judge exhibits a mild upward bias ($\Delta(L - H) = +0.20$), awarding 5/5 whenever a message cites specific numbers and respects constraints, whereas human raters assign 4/5 when phrasing combines two actions or omits a secondary context detail.
- **Notable Disagreements:** Three cases illustrate instructive divergence between human and LLM evaluators:
  1. In `case_003` (`u19`, three deadlines today and isolation comment *"loneliness is a cruel thing"*), the LLM Judge awarded 5/5 across all dimensions ($\mu = 5.00$), whereas the human rater scored Context Awareness as 4/5 and Clarity as 4/5 ($\mu = 4.33$, $|\Delta| = 0.67$), noting that the 3-clause message was slightly long and did not cite the exact $5/5$ stress rating.
  2. In `case_002` (`u08`, 2.8h sleep and $5/5$ stress on a CS project), the LLM Judge awarded 5/5 across all dimensions because the message cited both the 2.8h sleep figure and the three upcoming deadlines. The human rater scored Clarity as 4/5 ($\mu = 4.67$), noting that combining a 10-minute hallway stretch suggestion *and* an early-bedtime nudge in the same message asks two actions of an exhausted student.
  3. In `case_011` (`u08`, synthetic mask hiding sleep, activity, and stress), the LLM Judge awarded 5/5 on Goal Alignment for appropriately asking a single clarifying question under missing data, while the human rater gave 4/5 on Goal Alignment ($\mu = 4.67$) because the question focused on energy and study breaks without explicitly naming the student's posture-reset goal.

7. Ablation Study

Table 3 and Figures 2 to 4 present our six controlled ablation comparisons:

1. Multi-Agent vs. Single-Agent (`full` vs. `single`). While the single-agent baseline has access to the exact same sensing tools, memory store, and skill library and runs with half the LLM calls (3.0 vs. 6.0 calls/case, 12.4k vs. 15.6k tokens, 13.0s vs. 35.6s), its decision accuracy drops from 100.0% (`full`) to 95.0% (`single`, 75.0% on missing-data cases where it fails to ask a clarifying question on one masked case). Separating the *Planner*, *Context Analyst*, *Writer*, and *Critic* enforces explicit missing-data hypothesis checking before drafting.

2. Impact of Reflection and Reflection Depth ($K_r \in \{0, 1, 2, 3\}$). Disabling the Critic (`no_refl`) reduces LLM

Table 3. Controlled ablation study across nine system configurations (`configs/ablations.toml`). We compare multi-agent vs. single-agent design (`full` vs. `single`), planning on/off and depth $K_p \in \{0, 1, 2, 3\}$, reflection on/off and depth $K_r \in \{0, 1, 2, 3\}$, and model choice (`full` vs. `modelB`). Depth-3 configurations (`plan3`, `refl3`) and `modelB` are evaluated on a 10-case stratified subset covering all five categories and a complete 3-step preference episode.

Configuration	Succ. (%)	Dec. Acc.	Pref. Adh.	Context	Goal	Clarity	Mean	Calls	Tokens
Full (P=2, R=2)	**100**	**100.0%**	**100%**	**5.00**	**5.00**	**5.00**	**5.00**	**6.0**	**15573**
Single-Agent	100	95.0%	100%	5.00	5.00	5.00	5.00	3.0	12408
No Planning	100	85.0%	100%	4.75	4.81	4.94	4.83	4.7	10642
Plan Depth=1	100	90.0%	100%	4.73	4.87	5.00	4.87	5.8	14936
Plan Depth=3	100	100.0%	100%	5.00	5.00	5.00	5.00	6.1	15227
No Reflection	100	95.0%	100%	5.00	5.00	5.00	5.00	5.2	13817
Refl Depth=1	100	100.0%	100%	5.00	5.00	5.00	5.00	5.8	14606
Refl Depth=3	100	100.0%	100%	5.00	5.00	5.00	5.00	5.8	14391
Model B (Flash-Lite)	100	100.0%	100%	5.00	4.75	5.00	4.92	7.6	22736

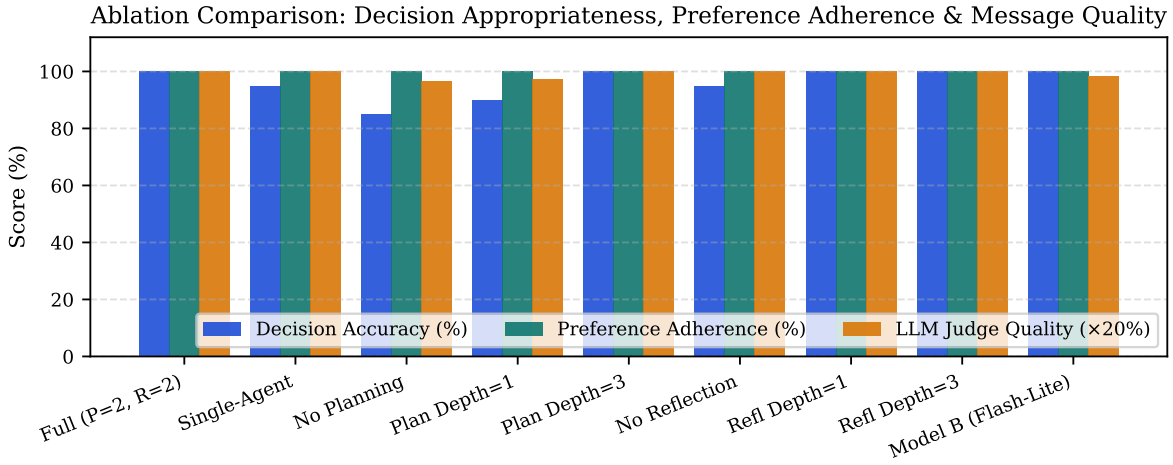


Figure 2. Ablation comparison across all nine configurations on Decision Appropriateness (%), Multi-Step Preference Adherence (%), and LLM Judge Message Quality (scaled to 100%).

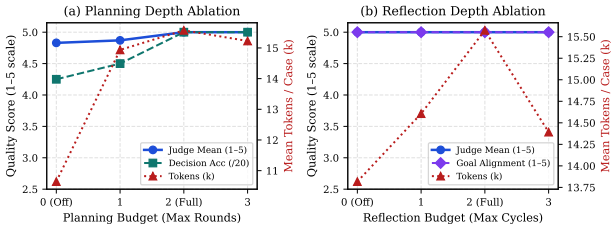


Figure 3. Quality vs. token cost trade-off curves as a function of (a) maximum planning rounds $K_p \in \{0, 1, 2, 3\}$ and (b) maximum reflection cycles $K_r \in \{0, 1, 2, 3\}$.

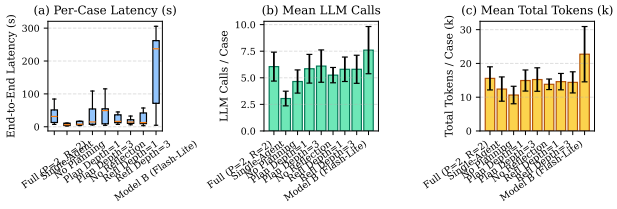


Figure 4. End-to-end per-case efficiency across configurations: (a) latency distribution (seconds), (b) mean LLM calls ± SD, and (c) mean total tokens ± SD (in thousands).

calls (5.2 vs. 6.0) and tokens (13.8k vs. 15.6k), but decision accuracy drops to 95.0% (over-messaging on a `typical_doing_fine` case) and removes the audit loop that catches subtle word-choice and length violations (for instance, in `full case_018`, the Critic caught an idiomatic "running on very little sleep" and triggered a revision). Increasing reflection budget from $K_r = 0$ to $K_r \in \{1, 2, 3\}$ (Figure 3(b)) restores 100.0% decision accuracy, while $K_r = 3$ (`refl3`) uses on average 1.0–1.13 reflection rounds because most drafts pass the Critic's audit within 1–2 cycles.

3. Impact of Planning and Planning Depth ($K_p \in \{0, 1, 2, 3\}$). Removing the Planner (`no_plan`) causes the largest degradation across all ablations: decision accuracy drops from 100.0% to 85.0% (66.7% on `typical_doing_fine`, 75.0% on `missing_data`, and 83.3% on `preference_update`), and LLM Judge quality drops from 5.00 to 4.83 (4.56 on preference updates). Comparing $K_p = 1$ (`plan1`, 90.0% decision accuracy, 4.87 judge mean) against $K_p = 2$ (`full`, 100.0% decision accuracy, 5.00 judge mean) and $K_p = 3$ (`plan3`, 100.0% decision accuracy, 5.00 judge mean) in Figure 3(a) shows that $K_p = 2$ is the sweet spot: Round 1 formulates initial hypotheses and Round 2 finalizes the decision after inspecting the Analyst's tool evidence, whereas $K_p = 1$ forces premature finalization before all modalities are cross-checked.

4. Model Choice (`full` vs. `modelB`). Evaluating the identical multi-agent architecture on our alternative model configuration (`modelB`, pairing `gemini-3.1-flash-lite` and `gemma-4-26b-a4b-it`) preserves 100.0% decision accuracy and 100% preference adherence with a $4.92/5.00$ judge mean (5.00 Context Awareness, 4.75 Goal Alignment, 5.00 Clarity), while requiring more schema-repair and reflection turns (7.6 calls and 22.7k tokens/case), confirming that structured multi-agent scaffolding and Pydantic schema repair transfer across model families.

8. Limitations and Conclusion

Limitations. First, because StudentLife lacks ground-truth coaching goals and conversational responses, our student profiles and multi-step feedback scripts are simulated (though deterministically grounded in pre-term psychometric surveys). Second, smartphone-inferred sleep relies on screen-dark and lock/charge heuristics that miss nights when a phone is left unlocked or off-charger, creating natural sensing gaps. Third, our automated judge uses a model from the same vendor family, which we mitigate via temperature-0 anchored rubrics and blind dual human evaluation.

Conclusion. We presented a modular, five-agent stress coach that combines leak-free mobile-sensing tools, iterative planning, skill-guided writing, self-reflection, and persistent preference memory. Our experiments and ablations on StudentLife confirm that structured multi-agent decomposition and explicit critique loops substantially improve coaching quality and longitudinal preference adherence.

Impact Statement

Automated mental-health and stress-coaching systems carry sensitive ethical responsibilities regarding student privacy, non-clinical boundaries, and crisis safety. Passive smartphone sensing captures intimate behavioral patterns (sleep schedules, conversation frequency, and emotional self-reports); in any real-world deployment, all feature extraction and tool queries must execute locally or under strict user-controlled consent with immediate data deletion rights. Furthermore, an AI stress coach is *not* a licensed clinician and must never diagnose psychiatric conditions (such as major depressive disorder from PHQ-9 scores) or replace professional care. Our system enforces strict non-diagnostic language rules in both the Message Writer and Critic prompts, and automatically routes indicators of acute distress (e.g., maximum EMA stress paired with depressive or self-harm survey signals, as in `case_019`) to curated human support resources, including Dartmouth College Health Service Counseling (`Dick's House`) and the 24/7 988 Suicide & Crisis Lifeline.

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Algorithm 1 Multi-Agent Stress Coaching Pipeline

Input: Case (u, t_{dec}, M), Config (K_p, K_r, flags), Store $\mathcal{D}$, Memory $\mathcal{M}$
$V \leftarrow \mathcal{D}.\text{get_view}(u, t_{\text{dec}}, M)$; $\mathcal{T} \leftarrow \text{make_tools}(V)$
$m_{\text{ctx}} \leftarrow \mathcal{M}.\text{format}(u, \text{enabled} = \text{flags.mem}, k = 5)$
$E \leftarrow [\,]$; $P \leftarrow [\,]$; $d^* \leftarrow \text{null}$
if flags.planning **and** $K_p \geq 1$ **then**
  for $r = 1$ **to** K_p **do**
    $p_r \leftarrow \textsc{Planner}(u, t_{\text{dec}}, m_{\text{ctx}}, P, E, r, K_p)$
    $P \leftarrow P \cup p_r.\text{plan_steps}$; $d^* \leftarrow p_r$
    if $E = \emptyset$ **or** $p_r.\text{status} = \texttt{gather_more}$ **then**
      $e_r \leftarrow \textsc{Analyst}(u, t_{\text{dec}}, p_r.\text{plan_steps}, \mathcal{T})$
      $E.\text{append}(e_r)$
    end if
    if $p_r.\text{status} = \texttt{finalize}$ **and** $E \neq \emptyset$ **then**
      break
    end if
  end for
else
  $E.\text{append}(\textsc{Analyst}(u, t_{\text{dec}}, \emptyset, \mathcal{T}))$
end if
if $d^* \neq \text{null}$ **and** $d^*.\text{decision} = \texttt{none}$ **then**
  $y \leftarrow (\texttt{none}, \text{null}, \text{null})$
else
  $S \leftarrow \text{load_skill}(d^*.\text{skill})$ **if** flags.skills **else** null
  $w \leftarrow \textsc{Writer}(u, m_{\text{ctx}}, E, d^*, S)$
  if $w.\text{decision} \in \{\texttt{suggest}, \texttt{ask}\}$ **and** flags.refl **then**
    for $c = 1$ **to** K_r **do**
      $\text{rev} \leftarrow \textsc{Critic}(u, m_{\text{ctx}}, E, d^*, w, c, K_r)$
      if $\text{rev.verdict} = \texttt{pass}$ **then**
        break
      end if
      $w \leftarrow \textsc{Writer}(u, m_{\text{ctx}}, E, d^*, S, w.\text{msg}, \text{rev})$
    end for
  end if
  $y \leftarrow (w.\text{decision}, w.\text{skill}, w.\text{msg})$
end if
$\Delta_{\text{pref}} \leftarrow \textsc{MemoryMgr}(u, y, \text{scripted_feedback})$
$\mathcal{M}.\text{record_episode}(u, t_{\text{dec}}, y, \Delta_{\text{pref}})$
Return: JSONL record $(y, P, E, \Delta_{\text{pref}}, \text{trace})$

A. LLM Judge Prompt and Anchored Evaluation Rubric

Both the independent LLM Judge (`configs/judge.toml`, $T = 0.0$) and the blind human evaluators used the following three-criterion anchored rubric (`prompts/judge.md`):

- **Context Awareness (1–5):**
  - **5 (Excellent):** Specifically and accurately grounds the message in the student's current data (recent sleep hours, stress trend, imminent deadlines, term week, or explicitly acknowledges missing sensing/EMA data when asking a clarifying question) with zero hallucinated facts.
  - **3 (Adequate):** Mentions general student context (e.g., "busy week" or "feeling stressed") broadly consistent with the data, but misses key specific signals or feels generic.
  - **1 (Poor):** Contradicts available evidence, hallucinates numbers/events not in the data, or ignores missing-data uncertainty by making unfounded assumptions.
- **Goal Alignment & Preference Compliance (1–5):**
  - **5 (Excellent):** Directly advances one of the student's stated goals AND strictly respects all baseline preferences, constraints, and learned feedback from prior episodes (including avoided activities, preferred settings, and length limits).
  - **3 (Adequate):** Relevant to general student well-being and mostly aligns with goals, with minor omissions of preferences.
  - **1 (Poor):** Ignores the student's goals OR violates a stated constraint or learned preference from past feedback (e.g., suggesting an activity the student explicitly rejected).
- **Clarity, Actionability & Tone (1–5):**
  - **5 (Excellent):** Clear, warm, non-judgmental, concise (2–3 sentences), and offers a concrete, immediately manageable step (or a single low-burden question when `decision == "ask"`).
  - **3 (Adequate):** Understandable and polite, but slightly wordy, vague, or packs too many suggestions at once.
  - **1 (Poor):** Confusing, overly clinical/robotic, judgmental, or overwhelming.

B. Modular Coaching Skills Catalog

Table 4. Summary of the eight on-demand Markdown coaching skills (`skills/*.md`) available to the Planner and Message Writer.

Skill Name	Target Decision	Primary Trigger & Contraindications
`micro_study_break`	`suggest`	High deadline density or acute stress during study blocks; contraindicated when severe sleep debt is the dominant issue.
`sleep_hygiene_nudge`	`suggest`	Inferred sleep < 6.0 h or declining vs. baseline; avoids lecturing or rigid early-wake advice for night owls.
`gentle_movement`	`suggest`	Prolonged sedentary coding/study periods with moderate stress; strictly avoided if student rejected exercise/walking.
`social_reconnection`	`suggest`	Low conversation minutes paired with high loneliness; contraindicated when student prefers solo recovery during exams.
`deadline_chunking`	`suggest`	≥ 2 deadlines within 72 hours causing task paralysis; breaks work into one 25–45 minute concrete block.
`campus_resource_referral`	`suggest`	Severe stress ($5/5$), distressed comments, or elevated PHQ-9/PSS; refers to Dick's House Counseling, Academic Skills Center, and 988 Lifeline.
`clarifying_question`	`ask`	Key sensing and EMA modalities return `status="missing"` (natural gap or synthetic mask); asks one brief check-in question.
`positive_reinforcement`	`suggest`	Student is maintaining healthy sleep/activity during moderate load and welcomes brief encouragement.